GEORGINA ADAMS

Birmingham, UK
07925672077

georgie.adams@live.co.uk

DESCRIPTION

I recently completed a master's degree in Bioinformatics at the University of Birmingham, building on my biomedical science background. I am now seeking opportunities to apply this expertise in a professional role. My academic experience includes computational biology, statistical machine learning and advanced omics technologies, including data analysis from next-generation sequencing technologies, single-cell and spatial transcriptomics. I have used and implemented R and Python throughout my degree, developing a strong technical skill set in both languages alongside a solid foundation in quantitative data analysis and statistical testing methods. My group research project is currently being developed toward co-authored publication.

Technical Skills

Programming: R, Python, Bash

Bioinformatics: Multi-omics data integration (including SGCCA, WGCNA), RNA-seq analysis (single-cell and spatial transcriptomics), biological data preprocessing and quality control, differential expression analysis, pathway and enrichment analysis, cell-type annotation and marker gene curation, cross-platform data integration

Machine Learning: Supervised and unsupervised learning, feature selection, model evaluation, clustering and classification methods.

Data Visualisation: ggplot2, matplotlib, seaborn, patchwork/cowplot for multi-panel figures, publication quality figures, exploratory data analysis

Data Handling: Large-scale biological datasets, genomic, transcriptomic and proteomic data, reproducible workflows and statistical analysis, high-performance computing.

EMPLOYMENT HISTORY

Care Assistant, Sanctuary Care.
From: 11/2019 **To:** Present.

GEORGINA ADAMS
Birmingham, UK
07925672077
georgie.adams@live.co.uk

I have worked as a bank member of staff in care homes throughout the course of my studies since 2019, where I have worked has depended on where I lived at the time based on it being termtime/summer break.

Post Graduate Teaching Assistant, University of Birmingham.

Supporting the delivery of a Masters-level module on the foundations of computing practices in health data science, covering programming in R, Python and SQL, relational databases, and version control, with applications to electronic health record systems and health data analytics.

EDUCATION

MsC Bioinformatics, University of Birmingham, Grade: TBC
From: 09/2025 To: Present

BsC Biomedical Science, Birmingham City University, Grade: First Class.
From: 09/2022 **To:** 06/2025.

CertHE Adult Nursing, Birmingham City University, Grade: Pass.
From: 01/2021 **To:** 12/2021

A-Level Education

Worcester Sixth Form College
From: 09/2017 To: 06/2019

Subject	Grade
Biology	C
Psychology	B
Health and Social Care	A*

GCSEs Education

9 GCSEs Grades A-C including English and Maths.

GEORGINA ADAMS
Birmingham, UK
07925672077
georgie.adams@live.co.uk

Projects

MSc Dissertation

Context-Dependent Reliability of Cell-Type Markers in Single-Cell and Spatial Transcriptomics
*(*Undertaken at Birmingham Tissue Analytics*)*

University of Birmingham | [May/26] – [Aug/26]

- Curated and validated a 42-marker panel for five immune/endothelial liver lineages from singlecell RNA-seq data, using combined statistical (log2FC, AUC) and visual validation criteria
- Independently assessed cross-platform marker transferability to CosMx spatial transcriptomics (1,000- and 6,000-gene panels), identifying panel coverage and detection sensitivity as primary constraints on transfer, rather than marker quality
- Built a full R/Seurat analysis pipeline (QC, clustering, differential expression, cross-platform statistical comparison) and produced publication-style figures using ggplot2, patchwork and custom visualisation tools

MSc Group Research Project
Transcriptomic analysis of seasonal gene expression in *Quercus robur*

UNIVERSITY OF BIRMINGHAM | [Oct/26] – [May/26]

I conducted a comprehensive RNA-seq differential expression analysis investigating the molecular basis of flowering regulation in pedunculate oak (quercus robur) under ambient and elevated co2 conditions, using data generated at the Birmingham institute of forest research (BiFor) face facility. I was responsible for most of the bioinformatics pipeline. This work is currently being prepared for publication. This involved the following:

- Preprocessing of raw sequencing data using Fastqc, Trimmomatic, hisat2, Samtools, and HTseq-count differential expression analysis in r using deseq2, including seasonal pairwise comparisons, eco_2 vs ambient within-month comparisons, and juvenile vs adult developmental comparisons

- Generation of all visualisations including volcano plots, upset plots, heatmaps, and venn diagrams compilation of priority candidate gene lists for downstream go enrichment and blast annotation.

Tools and languages: R, DEseq2, ggplot2, pheatmap, upsetr, ggvenn, hisat2, samtools, htseq, fastqc, trimmomatic, bluebear hpc, GitHub

MSc Coursework: Mathematical & Statistical Analysis using R

GEORGINA ADAMS
Birmingham, UK
07925672077
georgie.adams@live.co.uk

Applied statistical analysis and reproducible R workflow

UNIVERSITY OF BIRMINGHAM | [Oct/25] – [DEC/25]

- Built a reproducible R workflow to analyse probability distributions, statistical inference, and mathematical models, applying core techniques used in bioinformatics data analysis.
- Designed modular, reusable R functions to streamline statistical computation, improve code clarity, and support scalable, maintainable analysis pipelines.
- Conducted hypothesis testing, distribution fitting, and numerical analysis, presenting results through structured, publication-style visualisations.

BSc Dissertation

Computational Identification of Bioportides Targeting EFCAB9 for Non-Hormonal Male Contraception

- Built a bioinformatics workflow to identify the sperm-specific target protein EFCAB9, isolate its key binding motif, and design two candidate peptide therapeutics (bioportides), then validated their efficacy, safety, and structural stability entirely in silico.
- Skills gained: database mining and integration, target identification and validation, peptide/protein sequence design, interpretation of machine learning prediction outputs, protein structure prediction and visualisation, in silico safety/efficacy evaluation.

Tools used: DisGeNET, UniProt, Human Protein Atlas (HPA), Protein Data Bank (PDB), Complex Portal, MLCPP 2.0, InterPro, PepCalc, Immune Epitope Database (IEDB), AlphaFold, ChimeraX.